Task 1.3

1. Project Management Plan (PMP)

Project: Preparing for Influenza Season
Role: Senior Data Analyst

1. Stakeholder Communication Plan

1.1 Stakeholder Identification & Analysis

Stakeholder	Role & Decision Authority	Data Literacy	Primary Concerns (from brief & quotes)
Staffing Agency Administrators	Own staffing allocation decisions; approve final staffing plan	Medium–High	Efficient use of limited staff, state-level differences, avoiding over/understaffing
Hospitals & Clinics	Consumers of staffing decisions; operational feedback	Medium	Adequate staffing during peak flu periods; vulnerable patient care
Medical Agency Frontline Staff (nurses, PAs, doctors)	Execution-level; no allocation authority	Low–Medium	Deployment timing, workload intensity, fairness
Influenza Patients	No decision authority	Low	Quality and availability of care

1.2 Comunication Plan (Cadence & Escalation)

Stakeholder	Cadence	Format	Owner
Staffing Agency Administrators	Bi-weekly	Structured interim report + review meeting	Analytics Lead
Hospitals & Clinics	Monthly (read-only)	Executive summary deck	Analytics Lead
Frontline Staff	End of project	Summary briefing	Agency Admins
Mentor / Reviewer (implicit in deliverables)	Per achievement	Formal submission	Analyst

2. Schedule & Milestone

Preparing & Analysing Data - Achievement 1

Milestone	Objective	Deliverable	Acceptance conditions
1. Requirements & Questions	Translate business needs into answerable questions	Requirements & Data Questions Document	Questions map directly to objectives & requirements
2. Research Design	Define analytical approach & hypothesis	Research Design & Hypothesis Doc	Hypothesis is testable with available data
3. Data Sourcing	Identify datasets & constraints	Data Inventory & Limitations Assessment	≥2 datasets from different sources documented
4. Data Profiling	Assess data structure & integrity	Data Profiles (per dataset)	Data types, issues, summary stats included
5. Data Quality	Evaluate completeness, uniqueness, timeliness	Data Quality Measures Report	Quality issues explicitly documented
6. Data Integration	Combine datasets into unified analytical base	Integrated Analytical Dataset	Successful join with transformations explained
7. Statistical Analysis	Identify relationships & variability	Statistical Analysis Output	Variance, SD, ≥1 correlation identified

8. Hypothesis Testing	Test defined hypothesis	Hypothesis Test Results	Test conducted and interpreted
9. Interim Report	Consolidate insights	Interim Analytical Report	Covers findings to date; review-ready

Data Visualisation & Storytelling - Achievement 2

Milestone	Objective	Deliverable	Acceptance conditions
10. Viz Strategy	Define visualisation purpose & standards	Viz Design Checklist	Checklist applied consistently
11. Comparative Charts	Show composition & comparison	Bar / Pie / Treemap (Tableau)	Aligned to checklist
12. Temporal Analysis	Examine seasonality & peaks	Time Series & Forecast Viz	Seasonal patterns clearly visible
13. Distribution Analysis	Understand variable spread	Histogram & Box Plot	Distribution insights articulated
14. Correlation Viz	Visualize relationships	Scatter / Bubble Chart	Correlation visually supported
15. Spatial Analysis	Show state-level differences	Choropleth / Spatial Map	Justification documented
16. Storytelling	Convert analysis into narrative	Tableau Storyboard	Narrative aligned to objectives
17. Final Presentation	Communicate decisions & actions	Recorded Stakeholder Presentation	Includes conclusions & next steps

3. Project Deliverables

Deliverable Register (Explicitly Referenced to Brief)

Milestone (from Schedule)	Deliverable	Purpose in Project	Brief Reference
Requirements & Questions Definition	Requirements & Data Questions Document	Translate business requirements into analysable questions that guide all downstream work	Business Requirements; Requirements; Preparing for Influenza Season – Questions to Guide the Analysis
Research Design & Hypothesis Formulation	Research Design & Hypothesis Document	Define analytical approach and establish a testable hypothesis aligned to staffing objectives	Exercise 1.3: Designing a Data Research Project; Analysis Criteria
Data Sourcing & Relevance Assessment	Data Inventory & Limitations Assessment	Identify usable datasets and explicitly document constraints that affect analysis validity	Data Sets; Requirements – Assess data limitations
Data Profiling & Integrity Review	Data Profiles (per dataset)	Understand data structure, quality, and necessary cleaning steps	Exercise 1.5: Data Profiling & Integrity
Data Quality Evaluation	Data Quality Measures Report	Assess completeness, uniqueness, and timeliness to determine analytical reliability	Exercise 1.6: Data Quality Measures
Data Transformation & Integration	Integrated Analytical Dataset	Create a unified dataset enabling cross-variable analysis	Exercise 1.7: Data Transformation & Integration
Descriptive Statistical Analysis	Statistical Analysis Outputs	Quantify variability and identify relationships between key variables	Exercise 1.8: Conducting Statistical Analyses
Hypothesis Testing	Hypothesis Testing Results	Statistically test assumptions related to influenza trends and vulnerable populations	Exercise 1.9: Statistical Hypothesis Testing
Interim Insight Consolidation	Interim Analytical Report	Synthesize findings from Achievement 1 into a reviewable checkpoint	Exercise 1.10: Consolidating Analytical Insights

Visual Design & Strategy	Visualisation Design Checklist	Ensure visual consistency and appropriateness for stakeholders	Exercise 2.1 & 2.2
Comparative & Compositional Analysis	Comparative Visualisations (bar, pie, treemap)	Compare state-level and population-level differences	Exercise 2.3: Composition & Comparison Charts
Temporal Analysis & Seasonality	Temporal Visualisations & Forecast	Identify flu season timing and peak periods across states	Exercise 2.4: Temporal Visualisations & Forecasting
Distribution Analysis	Histograms & Box Plots	Understand distribution of key variables such as deaths or population size	Exercise 2.5: Statistical Visualisations
Correlation Visualisation	Scatter / Bubble Charts	Visually support statistical relationships between variables	Exercise 2.6: Scatter Plots & Bubble Charts
Spatial Analysis	Spatial Visualisation (Choropleth Map)	Show geographic variation in influenza impact and vulnerable populations	Exercise 2.7: Spatial Analysis
Narrative Development	Tableau Storyboard	Convert analytical findings into a coherent, decision-oriented story	Exercise 2.9: Storytelling with Data Presentations
Stakeholder Communication	Final Recorded Presentation	Communicate conclusions, recommendations, and next steps	Exercise 2.10: Presenting Findings to Stakeholders

4. Audience Definitions

Audience	Role (Who they are)	Objective (Decision Owned)	Task (What they do with output)
Staffing Agency Administrators	Senior operational and planning leaders at the staffing agency; moderate–high analytical literacy	Decide when and how many staff to allocate per state during influenza season	Review insights, approve staffing plan, balance utilisation vs. risk
Hospitals & Clinics	Operational managers and clinical leads consuming temporary staff; medium data literacy	Ensure adequate staffing during peak influenza demand	Use timing and severity insights to plan capacity and request staff
Frontline Medical Staff (Nurses, PAs, Doctors)	Temporary clinical workforce deployed across states; low–medium data literacy	Understand deployment rationale and expectations	Prepare for seasonal workload and location assignments
Influenza Patients (incl. Vulnerable Populations)	End beneficiaries of staffing decisions; no analytical expertise	Indirect—improved access to care and outcomes	Not direct users of outputs

2. Clarifying & Funnelling Questions - Additional Context (Project Brief)

Clarifying Questions

	Clarifying Question	Business Decision Supported	Data Explicitly Available in Brief	Planned Analysis (Within Scope)
Q1	Which states have the highest influenza-related deaths among vulnerable populations?	Where to prioritise staffing allocation under limited supply	- CDC influenza deaths by state - US Census population by age	- Calculate influenza death rates by state - Overlay vulnerable age groups (≥65, <5) - Rank states into low / medium / high need
Q2	When do influenza cases peak in each state, and how consistent is timing year over year?	When to deploy staff to avoid under- or over-staffing	- CDC FluView lab test counts by state and time - Influenza deaths time series	- Identify peak week/month per state - Compare peak timing across years - Assess consistency of seasonal patterns
Q3	What is vaccination coverage among vulnerable populations by state or region?	Adjust staffing needs based on preventive coverage	- CDC survey of flu shot rates in children	- Compare child vaccination rates with influenza deaths by state

Funnelling Questions

	Funnelling Question	Parent Clarifying Question	Data Explicitly Available in Brief	Feasible Analysis
Q1	Do states with higher vaccination rates see proportionally fewer influenza deaths?	Q3	- Child flu shot survey data - Influenza deaths by state	- Correlation analysis between vaccination rates and death rates
Q2	Is influenza season synchronised across all states, or do peak months vary regionally?	Q2	- FluView lab test counts - Influenza deaths time series	- Identify and compare peak months by state - Group states by early/mid/late season
Q3	Do influenza deaths increase in parallel with higher vulnerable population concentration?	Q1	- Influenza deaths by state - Census age-based population data	- Compare death rates against vulnerable population size

3. Hypotheses based on data sets (project brief)

Summary

Column1	Hypothesis	Primary Decision Supported	Direct Staffing Impact
H1	**Age → Mortality**	Where severity risk is highest	Prioritise high-elderly states
H2	**Seasonality → Peaks**	When to deploy staff	Time-based allocation
H3	**Vaccination → Severity**	Where prevention reduces demand	Adjust staffing intensity

Hypothesis 1

Focus:	Relationship between age (independent variable) and influenza mortality.

IF the proportion of adults aged 65 years and older in a state is higher, **THEN** the state will exhibit a **higher influenza-related mortality rate**, holding geography and time constant

	Element	Dataset	Variable
Variable Mapping:	Independent variable	US Census Population Data	Population count or share age ≥65
	Dependent variable	CDC Influenza Deaths	Influenza deaths by state and year
	Unit of analysis	State–year	—

4. Data Wishlist

Age and Influenza Mortality

	Data Item	Purpose / Use	Source	Notes / Constraints
W1	**State-level population by age group**	Independent variable: quantify the size/proportion of elderly (≥65 years) in each state	US Census Bureau Population Data	Ensure consistency with year of influenza death data; use age bands as defined in dataset
W2	**State-level influenza deaths**	Dependent variable: measure influenza severity for mortality analysis	CDC Influenza Deaths Data	Historical counts by state; aligns with the same time period as population data